



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 11

Data Science and Financial Modelling: Applications and Cases (Equity)



STUDY NOTES • EDITION 1.0 • WEEK 11

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How to use these notes



These notes cover the whole of Week 11: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A asks what data can honestly tell us, and introduces the two statistical ideas — sampling noise and Bayes' theorem — that discipline every claim in the module. Part B opens the machinery: regression, the vocabulary, and the three families of machine learning. Part C covers where data comes from, which techniques exist, and how each input from Weeks 5 and 6 can be improved. Read the parts in order; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

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CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview



This module asks a question the earlier weeks could not: if a computer can evaluate a hundred variables at once, why does CAPM use three? It begins with what data can honestly tell us — sampling noise, and the Bayesian arithmetic of updating a belief — then opens the regression you have been using since Week 6 and shows it is a machine learning model. It closes by rebuilding each input of portfolio theory with better information.

One thread runs through all nine topics: **artificial intelligence is not replacing finance**. The financial questions and concepts stay the same. What changes is the speed, scale and depth at which they can be applied — and the honesty with which their uncertainty can be stated.

Learning objectives

By the end of this module, you will be able to:

- 1 **Explain** why classical models use few variables, and describe sampling noise and its consequences.
- 2 **Apply** Bayes' theorem in counts and interpret a linear regression as a machine learning model.
- 3 **Distinguish** the three families of machine learning and describe financial data sources.
- 4 **Explain** how expected returns, correlation and beta can each be improved by better estimation.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	Why three variables are not enough	A • Evidence	L37 V1	LO 1
2	How much can data actually tell us?	A • Evidence	L37 V2	LO 1
3	Bayes' theorem	A • Evidence	L37 V3	LO 2
4	Inside linear regression	B • Machinery	L37 V4	LO 2
5	The finance-machine learning dictionary	B • Machinery	L37 V5	LO 2
6	The three families of machine learning	B • Machinery	L37 V6	LO 3
7	Where financial data comes from	C • Application	L37 V7	LO 3
8	The ladder of techniques	C • Application	L37 V8	LO 3
9	Rebuilding the pyramid: better inputs	C • Application	L38 V1–V2	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Inferential statistics

Drawing conclusions about a large unseen population from a smaller observed sample.

Sample

The observed subset actually available — for instance sixty monthly returns.

Standard error

The usual amount by which an estimate fluctuates between samples.

Prior

What was believed before new evidence arrived.

Posterior

The updated belief after weighing the evidence.

Least squares

The method selecting the line that minimises the sum of squared residuals.

Multiple regression

Regression using several input variables at once.

Features

The machine learning name for independent variables.

Weights

The machine learning name for regression coefficients.

Supervised learning

Learning from past examples where the answer is known.

Clustering

Grouping items by similarity without being given categories.

Regularisation

Charging a cost for model complexity, to curb confidence in accidental patterns.

Unstructured data

News, reports, calls and images — information not naturally tabular.

API

An interface letting one programme request data from another system automatically.

Population

The complete but unobservable set of outcomes.

Estimate

A value calculated from a sample; the true value plus sampling noise.

Hypothesis test

A formal method for judging whether a pattern is real or could have arisen by chance.

Likelihood

How probable the evidence would be under each possible explanation.

Residual

The vertical distance from a data point to the regression line; the unexplained part.

R-squared

The proportion of the target's movement explained by the model.

Target, or label

The machine learning name for the dependent variable.

Training

Estimating the relationship between features and target from historical data.

Overfitting

Memorising historical data rather than learning a pattern that persists; data mining.

Unsupervised learning

Finding hidden structure in data with no answers attached.

Reinforcement learning

Learning a sequence of decisions through trial, reward and penalty.

Structured data

Information arranged in rows and columns.

Sentiment score

A number from -1 to $+1$ representing how positive or negative a text appears.

Heat map

A colour-coded grid displaying many correlations at once.

PART A • WHAT DATA CAN AND CANNOT TELL YOU

How much can a number be trusted?

1 Why Three Variables Are Not Enough

PART A • WHAT DATA CAN AND CANNOT TELL YOU • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

CAPM uses three inputs because 1960s computers could not manage more — not because more would not help.

Why this matters

This module does not replace the finance of Weeks 5 and 6. It asks why those models were built so small, and what becomes possible once the constraint that shaped them is removed.

Meera's question

Meera now helps manage ₹5,000 crore of other people's savings. One question is on her screen: should the fund invest another ₹500 crore in Reliance Industries? The committee meets at nine tomorrow. What return should she expect?

After Weeks 5 and 6 the answer comes quickly — use CAPM. Risk-free rate, beta, market risk premium. Three inputs, one answer. It is elegant, it contributed to Nobel Prize-winning work, and it has been taught for sixty years.

Now the uncomfortable question. Sitting at a desk with ₹500 crore of investors' savings at stake, would those three numbers honestly be the *only* things you examined?

What you would actually look at

Within a minute the list runs well past three: the price-to-earnings ratio, return on equity, debt levels, promoter holding, how much foreign and domestic institutions are buying, crude oil prices (Reliance refines oil), the rupee-dollar rate (it buys crude in dollars), the latest quarterly results, the tone of management on the earnings call, what analysts are saying, and the discussion on social media.

Nobody has said “artificial intelligence” yet, and the case for it is already made.

Why the founders used so few

Not because they were less intelligent, but because computers were slow and data was scarce. In 1952, when Markowitz developed portfolio theory, studying the relationships among even fifty shares could take months. In 1964, running a single regression could mean booking overnight time on a university mainframe.

Under those constraints, simplifying the world to a handful of variables was not laziness — it was **necessary**. Machine learning begins from the opposite question: why should useful information be discarded when a computer can evaluate all of it at once?

SAME QUESTION, DIFFERENT AMOUNT OF EVIDENCE

CAPM**INPUTS**

Risk-free rate, beta, market risk premium

NUMBER OF INPUTS

Three

OUTPUT

An expected return

WHY SO FEW

In 1964 a single regression meant overnight mainframe time

A MACHINE LEARNING MODEL**INPUTS**

Ratios, flows, commodity prices, currency, news tone

NUMBER OF INPUTS

Perhaps a hundred

OUTPUT

An expected return

WHAT CHANGED

Not the question — only the information used to answer it

*By the time trading begins, Meera's desk has processed only a **small share** of what could have been considered. Two analysts cannot read every filing; the risk team works on yesterday's data while the market generates new information every second.*

Figure 1.1 The question is unchanged. Only the richness of the answer differs.

KEY IDEA

Artificial intelligence is not replacing finance. The financial questions and concepts stay exactly the same — what changes is the **speed, scale and depth** at which they can be applied.

EXAM POINTER

If asked why classical models use few variables, give the historical constraint with a date — 1952 for Markowitz, 1964 for CAPM-era regressions — and state that the simplification was necessary rather than careless. Then contrast the machine learning premise: why discard information a computer can evaluate at once?

QUICK RECAP • TOPIC 1

- CAPM answers a real question with three inputs, and has done so for sixty years.
- A practitioner would examine far more: ratios, flows, commodities, currency, sentiment.
- The founders used few variables because computers were slow and data scarce, not from carelessness.
- Machine learning asks the opposite question — why discard useful information?
- The financial question is unchanged; only the scale of evidence differs.

2 How Much Can Data Actually Tell Us?

PART A • WHAT DATA CAN AND CANNOT TELL YOU • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Every number you have calculated in this course is an estimate equal to the truth plus sampling noise.

Why this matters

This is the most important corrective in the module. Weeks 5 to 8 treated computed figures as facts; this topic explains why they are not, and how much doubt to attach to each.

Population and sample

Inferential statistics is the process of drawing careful conclusions about something large and unseen from a smaller set of observed data. It is the foundation of econometrics, machine learning and every model in this module.

The **population** is the true behaviour of Reliance Industries: how the stock would perform, on average, across every possible month in which the company could exist. It cannot be observed. The **sample** is the sixty monthly returns actually sitting in your spreadsheet.

Therefore every number in this course came from a sample. The average return, the standard deviation, the beta from SLOPE — each is an **estimate** of the underlying truth, equal to the true value plus sampling noise.

How wrong can an estimate be?

Flip a fair coin ten times. You should not be surprised to see seven heads — that happens more often than once in every six such experiments. Would you conclude the coin lands heads 70% of the time? Of course not. Ten flips is a very small sample.

The **standard error** measures the usual amount by which an estimate fluctuates from one sample to another. It has one particularly useful property: it shrinks with the square root of the sample size, so **to halve the standard error you need roughly four times as much data**.

Why not simply collect older data? Because older observations may come from a very different company in a very different economy. The Reliance of 2010 is not the Reliance of today.

The habit worth keeping

Suppose 500 fund managers each make a prediction every year, and their success is settled by a fair coin. After five years, roughly **15** will have beaten the market every single year purely by chance, because the probability of five straight wins is 1 in 32. Each now has a five-year winning record and may speak with great confidence.

So whenever data appears to reveal a pattern, ask: **could pure chance have produced this?** That question is the basis of a *hypothesis test*.

REPLAYING FINANCIAL HISTORY • THE SAME STOCK, MANY DIFFERENT ANSWERS

THE TRUTH

Suppose the true expected return is 12% • we can never observe it

ONE FIVE-YEAR SAMPLE

Whatever history happened to deliver

AN UNLUCKY SAMPLE

Reports just 4%

A FAVOURABLE ONE

Reports 20%

*The stock is the same and the truth is the same. Only the **sampling noise** differs — and the width of that spread is what the standard error measures.*

Figure 2.1 Estimation noise. Two honest analysts can reach very different figures from the same stock.

COMMON PITFALL

Two analysts study the same stock. One uses 2015 to 2019 and reports 4%; the other uses 2019 to 2023 and reports 20%. Which found the true expected return? **Neither**. Both are working with noisy estimates from small samples. If that feels uncomfortable, that is the correct reaction.

EXAM POINTER

Define population and sample in the financial context, and state that every computed figure is an estimate. Give the square-root property of standard error precisely — four times the data to halve it — and note why older data does not solve the problem.

QUICK RECAP • TOPIC 2

- Inferential statistics draws conclusions about an unseen population from an observed sample.
- Every figure computed in this course is an estimate: the truth plus sampling noise.
- Standard error shrinks with the square root of sample size — four times the data to halve it.
- Older data may come from a company and economy that no longer exist.
- Of 500 coin-flipping managers, about 15 will show a five-year winning record by chance alone.

3 Bayes' Theorem

PART A • WHAT DATA CAN AND CANNOT TELL YOU • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

A five-year winning record raises the probability that a manager is genuinely skilled from 5% to only about 50%.

Why this matters

Bayes is the arithmetic of changing your mind, and it recurs throughout Weeks 12 and 13 — in credit models, in crash alarms, and anywhere a rare event must be detected.

The three components

If every number is an estimate surrounded by noise, then when new evidence arrives, how much should we change our mind? There is an answer roughly 260 years old.

The **prior** is what we believed before the evidence arrived. The **likelihood** is how probable the observed evidence would be under each possible explanation. The **posterior** is our updated belief — the prior corrected by the strength of the evidence.

In one line: **the new belief is the old belief corrected by how well the evidence fits.**

The star fund manager illusion

Of 500 fund managers, suppose 5% are genuinely skilled — that is 25 managers. Skilled managers beat the market in 9 years out of 10. The other 475 have no skill; their results are equivalent to a fair coin. Five years pass.

Among the 25 skilled managers, $25 \times 0.9^5 \approx 15$ beat the market in all five years. Among the 475 unskilled managers, $475 \times (1 \div 32) \approx 15$ also beat it in all five years — purely by luck.

The group with a five-year winning record therefore holds about 30 people: roughly 15 genuinely skilled and roughly 15 simply fortunate. So even after an impressive record, the probability that a manager is genuinely skilled is only about **50%**. The prior was 5%; the posterior is 50%. The evidence changes our belief significantly — but it does not create certainty.

Why a 90% accurate model can still be wrong five times in six

A risk model claims to identify market crashes with 90% accuracy. Crashes occur in about 2% of months. This month the model issues a warning. What is the probability a crash is actually coming?

Work with counts across 1,000 months. Crashes occur in 20 months; the model catches 90% of them, giving **18 correct warnings**. That leaves 980 calm months, and if the model falsely warns in 10% of them, **98 false warnings**. Total warnings = 116, of which only 18 precede an actual crash.

Probability of a crash given a warning = $18 \div 116 \approx 16\%$. The warning is wrong about five times out of six — because when an event is rare, false warnings vastly outnumber correct ones.

TWO BAYES CALCULATIONS, BOTH COUNTED RATHER THAN DERIVED

	The star manager	The crash alarm
Prior	5% of 500 managers are skilled — 25 people	Crashes occur in 2% of months — 20 in 1,000
Evidence	A five-year winning record	The model issues a warning
True positives	$25 \times 0.9^5 \approx 15$ skilled managers	90% of 20 = 18 correct warnings
False positives	$475 \times (1 \div 32) \approx 15$ lucky managers	10% of 980 = 98 false warnings
Total showing the evidence	About 30 people	116 warnings
Posterior	$15 \div 30 \approx 50\%$ genuinely skilled	$18 \div 116 \approx 16\%$ followed by a crash

Both calculations work the same way: count the true positives, count the false positives, and divide. **When the event is rare, the false positives dominate.**

Figure 3.1 Bayes in counts. No algebra is needed — only careful bookkeeping.

KEY IDEA

Do not ask only how impressive the evidence is. Ask **how easily luck could have produced the same evidence**. An impressive back test must be weighed against the many attractive-looking back tests that chance alone would generate.

EXAM POINTER

Work Bayes problems in *counts* rather than formulas — it is faster and far less error-prone. Name prior, likelihood and posterior. The examinable insight is that rare events produce mostly false alarms even from accurate models, which is why a warning should trigger investigation rather than automatic action.

QUICK RECAP • TOPIC 3

- Prior, likelihood, posterior: the new belief is the old one corrected by how well the evidence fits.
- A five-year winning record lifts the probability of skill from 5% to only about 50%.
- About 15 skilled and 15 lucky managers share the same impressive record.
- A 90% accurate crash model issuing a warning is right only about 16% of the time.
- When events are rare, false warnings vastly outnumber correct ones.

PART B • THE MACHINERY

Opening the tools you already use

4 Inside Linear Regression

PART B • THE MACHINERY • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

The *SLOPE* function you used in Week 6 is the ancestor of much of what is now called machine learning.

Why this matters

Opening the regression is what makes the rest of the module unintimidating. Once you see that beta is a fitted coefficient, every more advanced model becomes a variation rather than a mystery.

The line through the cloud

Collect sixty months of data. Put the Nifty's monthly return on the horizontal axis and Zomato's return in the same month on the vertical. Sixty months produce sixty points, and the picture already tells a story: when the market did well, Zomato generally did too — but not every time, and not always by the same amount.

Linear regression asks one question: which single straight line passes as close as possible to all sixty points at once?

The three components

Stock return = alpha + (beta × market return) + error term.

Alpha is the intercept — the stock's estimated return in a month when the market returns nothing. *Beta* is the slope: if it is 1.30, a 1% market rise implies roughly a 1.30% rise in the stock. The *error term*, or residual, is the vertical distance from a point to the line — the part of that month's return the market cannot explain.

This repeats a Week 5 lesson in new language. What the line explains is **systematic risk**; what sits in the error term is **unsystematic risk**, the part diversification removes.

Least squares, and where machine learning begins

Measure how far every point lies above or below a candidate line — the residuals. Square each, so positives and negatives cannot cancel and large errors count more. Sum across all sixty. Choose the line minimising that total. This is **least squares**, over two hundred years old and originally developed for tracking planets.

R-squared measures the proportion of movement the model explains: if R^2 is 0.45, roughly 45% of Zomato's movement is explained by the market and 55% by company-specific factors.

So far the line only *describes* a historical relationship. **The moment we apply that fitted relationship to future values, statistical estimation becomes machine learning.**

FROM FITTED LINE TO PREDICTION — AND THE ERROR THAT REMAINS

Step	Working	Result
The fitted model	Regression on sixty months of returns	alpha = 0.2%, beta = 1.3
Expected market move	Next month	+2%
Predicted stock return	$0.2\% + (1.3 \times 2\%) = 0.2\% + 2.6\%$	2.8%
Actual stock return	What happened	4%
The residual	$4\% - 2.8\%$	1.2% — unsystematic risk

Why stop at one input? Add momentum, valuation, size, profitability and ownership, and each coefficient becomes a **weight the model learns**. That is multiple regression — the first rung of the ladder.

Figure 4.1 A prediction and its error. The 1.2% gap is company-specific information the market did not explain.

COMMON PITFALL

Forgetting that three assumptions sit underneath a simple regression: that a straight line is the right shape, that the data-generating process is reasonably stable over time, and that the errors are random noise rather than a hidden pattern. Machine learning often relaxes the first — it still faces the other two.

EXAM POINTER

Name all three components and describe least squares in four steps: measure, square, sum, minimise. Explain why squaring is necessary. Then state the transition precisely — estimation becomes machine learning when the fitted relationship is applied to future values.

QUICK RECAP • TOPIC 4

- Regression finds the single line passing closest to all the data points at once.
- Stock return = alpha + (beta × market return) + error term.
- Least squares minimises the sum of squared residuals; squaring stops errors cancelling.
- R^2 of 0.45 means 45% of movement is explained and 55% is not.
- Applying a fitted relationship to future values is where machine learning begins.

5 The Finance–Machine Learning Dictionary

PART B • THE MACHINERY • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Most of the intimidation in this field comes from unfamiliar vocabulary rather than unfamiliar substance.

Why this matters

You have been practising data science for weeks under different names. This topic supplies the translation, and one entry in it — overfitting — recurs in every remaining week.

You have already done this

Look again at Topic 4: you took historical data, fitted a model to it, and used the model to predict. That is a direct description of machine learning. Linear regression is not merely related to it — it is one of the oldest and simplest members of the family.

The same connection appears elsewhere. Arranging 500 days of historical returns to estimate a worst-case loss is what an engineer would call a *data-driven model* — one where the rules are learned from data rather than written in advance. The rules deciding which shares enter the Nifty are what a computer scientist would call an *algorithm* — simply a recipe, a fixed sequence of steps.

The translation that matters most

One entry deserves special attention. **Overfitting** occurs when a model memorises the historical data instead of learning a pattern that will keep working. It may look extremely successful when tested on the same data used to build it, and perform poorly on new, unseen data.

In finance this has an older name: *data mining* — searching history until an attractive pattern appears. The two words describe the same failure.

Every Excel function has a more advanced relative

AVERAGE gives the sample mean, an estimate containing noise; its extension is a return-ranking model that accounts more carefully for that noise. STDEV measures spread; its extension is volatility forecasting. CORREL measures one pair; its extension handles correlations across many assets, updated continuously. SLOPE and INTERCEPT give beta and alpha; their extension is the same framework with many inputs at once. LINEST performs multiple regression; its extension is *regularised* regression, which controls complexity to reduce overfitting.

Note too that the standard errors a regression reports are exactly the estimation uncertainty from Topic 2 — they show how much alpha and beta may vary because of sampling noise.

THE SAME IDEA, TWO VOCABULARIES

In finance	In machine learning	Example
Dependent variable	Target, or label	The return of a share we want to predict
Independent variables	Features	Market return, P/E ratio, return on equity, oil prices, rates
Estimating a model	Training a model	Learning the input-output relationship from history
Coefficients	Weights	How strongly each feature influences the prediction
Predicted value	Prediction	The model's output
Error term, residual	Error	The part the model failed to explain
R-squared	Goodness of fit	How much of the target's movement the model explains
Data mining	Overfitting	Searching history until an attractive pattern appears

Read the table in both directions. A finance graduate already knows the right-hand column — **the substance was never unfamiliar, only the words.**

Figure 5.1 The translation table. The last row names the same failure twice.

KEY IDEA

Overfitting is the central danger of the whole module. A model that fits history perfectly has often learned the *noise* in that history — and noise, by definition, will not repeat. Week 13 examines this in detail.

EXAM POINTER

Give at least five translation pairs. Define overfitting precisely — memorising the data rather than learning a generalisable pattern — and note that the finance term for the same failure is data mining. That equivalence is the examinable point.

QUICK RECAP • TOPIC 5

- Linear regression is machine learning: fit a model to history, then predict from it.
- Dependent variable becomes target; independent variables become features; coefficients become weights.
- Estimating a model is training it; the residual is the error.
- Overfitting means memorising the data rather than learning a pattern that persists.
- Its finance name is data mining — searching history until something attractive appears.

6 The Three Families of Machine Learning

PART B • THE MACHINERY • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Identify which family a task belongs to and you will already think more clearly than most people discussing AI in finance.

Why this matters

This taxonomy organises everything in Weeks 12 and 13. Every technique introduced later belongs to one of these three families, and knowing which tells you what it needs and what it can deliver.

Supervised learning

Learning from past examples where the correct answer is **already known** — practice questions that come with an answer key. Show the machine the financial characteristics of a stock in a given month alongside the return it actually delivered, and let it study the relationship.

Your Week 6 beta calculation was supervised learning. As a general rule, **predicting a number or a category is a supervised task**.

Unsupervised learning

Learning from data with **no answers attached**; the machine must find hidden structure itself. Picture a librarian handed 10,000 unlabelled books and asked to arrange them into meaningful groups — nobody says what the categories should be.

Its most important task is **clustering**. Give the machine the daily returns of 200 stocks and, without being told their sectors, it may find that banking stocks form one group, IT companies another, pharmaceutical and consumer goods others. It discovers behavioural families from the data itself — and may reveal a company officially labelled IT that in fact trades like a consumer goods stock.

Reinforcement learning

Learning through **trial, error and reward** across a sequence of decisions. It works the way a person learns to ride a bicycle: nobody explains the physics in advance; the learner tries, makes mistakes, adjusts and improves.

In finance it can decide how to spread a large purchase order across the trading day, learning which sequence of decisions works best under changing conditions. It is examined in Week 13.

FOUR TASKS, THREE FAMILIES • THE RULE OF THUMB AT THE BOTTOM

Task	Family	Why
Predict next quarter's beta for 80 stocks	Supervised	A numerical value predicted from past examples
Group those 80 stocks into behavioural families	Unsupervised	The groups must be discovered without given categories
Spread a ₹40 crore buy order across the day	Reinforcement	A sequence of decisions learned through feedback and reward
Identify which stocks may cut their dividend	Supervised	Past examples with known outcomes predict a category

The rule of thumb. A number or category to predict means supervised. Hidden structure to discover means unsupervised. A sequence of decisions learned through rewards means reinforcement.

Figure 6.1 The three families. Classifying the task correctly is the first analytical step.

REAL-WORLD EXAMPLE

Clustering can be genuinely revealing rather than merely tidy. A model given only return data may place a company in a different group from its official sector classification — which is useful information, because a portfolio believed to be diversified across sectors may in fact hold several stocks that behave identically.

EXAM POINTER

Name all three families with what distinguishes each, then apply the rule of thumb to a stated task. The discriminating detail for supervised learning is that the training data comes with known answers; for unsupervised, that no categories are supplied at all.

QUICK RECAP • TOPIC 6

- Supervised learning uses past examples where the correct answer is known.
- Unsupervised learning finds hidden structure in data with no answers attached.
- Clustering groups assets by behaviour and may contradict official sector labels.
- Reinforcement learning learns a sequence of decisions through trial, reward and penalty.
- A number or category means supervised; structure means unsupervised; a sequence means reinforcement.

PART C • DATA, TECHNIQUES AND BETTER INPUTS

From data to a better decision

7 Where Financial Data Comes From

PART C • DATA, TECHNIQUES AND BETTER INPUTS • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Nobody gathers a hundred signals by hand. Software collects, cleans and combines them on a schedule — and can now read documents too.

Why this matters

A fair objection to the whole module is that Meera has two analysts, not two hundred. This topic answers it, and introduces the unstructured data that makes the machine's advantage decisive.

Who collects it all

The encouraging answer is that this problem has largely been solved. Prices and volumes stream directly from exchanges. Company information such as profits, debt and margins sits in standard databases. Interest rates, crude oil prices and exchange rates arrive through public feeds. Software collects, cleans and combines all of it into one organised table on a fixed schedule with very little human involvement.

Four practical terms. A **CSV file** is a table saved as plain text — an Excel worksheet without formatting or formulas. A **database** stores millions of rows and answers questions within seconds. **SQL** is how analysts talk to a database. An **API** is a service counter letting one programme request information from another system automatically.

The practical boundary: Excel opens a CSV easily but cannot comfortably manage 50 million rows. That is the line between spreadsheets and databases.

Structured and unstructured data

Structured data is arranged in rows and columns: prices, ratios, volumes. **Unstructured data** does not naturally fit: news articles, annual reports, earnings call recordings, social media posts, satellite photographs. Historically the first was machine-readable and the second interpretable only by people.

Modern language models have changed that, converting words, audio and images into numbers analytical models can use.

Sentiment, and the wider frontier

A **sentiment score** represents how positive or negative a piece of text appears, typically on a scale from **-1 to +1**. A headline saying profits exceeded expectations might score +0.8; one saying margins are under pressure might score -0.5. The machine can calculate thousands of such scores and average them into one value per stock, updated through the day for every stock at once. A human research team may struggle to monitor even ten companies continuously.

Beyond headlines: a machine can listen to an earnings call and count words suggesting hesitation — “challenging”, “uncertain”, “we hope”. Managers under pressure may change how they speak before the numbers reveal the problem. Analysts have used satellite photographs of shop car parks to estimate sales before results, and studied shadows on oil storage tanks to gauge how full they are.

TWO KINDS OF DATA • AND WHAT CHANGED

STRUCTURED**DEFINITION**

Information arranged in rows and columns

EXAMPLES

Prices, financial ratios, trading volumes

HISTORICALLY

Machine-readable

WHERE IT LIVES

CSV files and databases, queried with SQL

UNSTRUCTURED**DEFINITION**

Information that does not fit rows and columns

EXAMPLES

News, annual reports, earnings calls, social posts, satellite images

HISTORICALLY

Interpretable only by people

WHERE IT LIVES

Documents, audio, images — now readable by language models

*This is the real sense in which a machine observes more of the financial world than any human team — **not because it is more intelligent, but because it can process 50,000 documents before breakfast.***

Figure 7.1 The two data types. The right-hand column is what recently became usable.

KEY IDEA

The machine's advantage is not superior judgement. It is **coverage** — the ability to attend to everything at once, continuously, without fatigue. That is a different claim from intelligence, and a more defensible one.

EXAM POINTER

Define all four data terms — CSV, database, SQL, API — and give the Excel boundary. Distinguish structured from unstructured with examples of each, and state the sentiment score range of -1 to +1 with what the endpoints mean.

QUICK RECAP • TOPIC 7

- Software collects, cleans and combines financial data automatically on a schedule.
- CSV, database, SQL and API are the four practical terms; Excel cannot handle 50 million rows.
- Structured data fits rows and columns; unstructured data — news, calls, images — does not.
- Sentiment scores run from -1 to +1 and can be computed for every stock continuously.
- The machine's advantage is coverage, not intelligence.

8 The Ladder of Techniques

PART C • DATA, TECHNIQUES AND BETTER INPUTS • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

A short ladder of techniques covers most of what is actually done in finance — and each rung appears where its financial problem does.

Why this matters

This topic maps the rest of the programme. It also introduces regularisation, which is the direct defence against the overfitting named in Topic 5.

The ladder

Beneath the three families sits a short ladder of techniques. Each is introduced where the relevant financial problem appears rather than as abstract method.

Why regularisation matters

Give an ordinary regression 50 noisy inputs and it may confidently identify patterns that appeared only by chance — claiming, say, that stocks whose names begin with the letter R performed well last year, and assigning that a high predictive weight.

Regularisation teaches the model humility by charging a cost for every unit of complexity or confidence added. It discourages excessive confidence in weak or accidental patterns, so the model keeps the inputs that genuinely contribute and reduces the importance of those that do not earn their place.

A story to set expectations

A famously secretive American investment fund employed mathematicians and code breakers rather than business school graduates, and over several decades earned extraordinary returns — by searching enormous amounts of data for faint statistical patterns and trading them with strict discipline.

The first lesson is that machines *can* identify genuine opportunities in financial markets. The second, more important lesson is that those opportunities are usually **weak and difficult to detect**. They are more like a biased coin flipped millions of times than a crystal ball. Keep that honesty throughout.

THE LADDER • EACH RUNG APPEARS WHERE ITS FINANCIAL PROBLEM DOES

Introduced	Technique	What it does
This week	Multiple regression	Explains a stock's return using many variables together — momentum, valuation, size, volatility, sentiment
This week	Regularisation	Charges a cost for complexity, curbing confidence in accidental patterns
This week	Clustering	Groups stocks into behavioural families discovered from the data itself
Week 12 • bonds	Principal component analysis	Reduces many moving variables into a few underlying drivers
Week 12 • bonds	Classification	Predicts a category rather than a number — default or not
Week 12 • bonds	Maximum likelihood estimation	Selects the model settings that make the observed data most probable
Week 13 • derivatives	Trees, forests, simulation, reinforcement learning	Handle the sharp bends and non-linear relationships in derivative payoffs

*Machines can find genuine opportunities. But those opportunities are usually **weak and hard to detect** — more like a biased coin flipped millions of times than a crystal ball.*

Figure 8.1 The ladder of techniques across the three data science modules.

COMMON PITFALL

Expecting a model to be right most of the time. A genuine statistical edge in markets may be a fraction of a percentage point, exploited across thousands of trades. A model that appears highly accurate is far more likely to be overfitted than exceptional.

EXAM POINTER

Name the techniques and the module in which each appears. Explain regularisation in terms of its purpose — charging a cost for complexity to prevent confident conclusions from accidental patterns — rather than its mathematics.

QUICK RECAP • TOPIC 8

- Multiple regression, regularisation and clustering are introduced this week.
- Principal component analysis, classification and maximum likelihood follow in Week 12.
- Trees, forests, simulation and reinforcement learning arrive in Week 13.
- Regularisation charges a cost for complexity, curbing confidence in accidental patterns.
- Real statistical edges are weak and hard to detect — a biased coin, not a crystal ball.

9 Rebuilding the Pyramid: Better Inputs

PART C • DATA, TECHNIQUES AND BETTER INPUTS • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Markowitz invented portfolio optimisation and did not use it for his own savings. The mathematics was never the problem — the estimates were.

Why this matters

The closing topic returns to Weeks 5 and 6 and improves each input in turn. It is the payoff of the entire module.

What Markowitz did with his own money

Harry Markowitz invented portfolio optimisation and won a Nobel Prize for it. Late in life an interviewer asked how he invested his own retirement savings. Surely the inventor of the optimiser used the optimiser.

He did not. He simply split his savings equally between stocks and bonds, so that whatever the markets did, he would not regret his decision too much.

This is not a criticism. Markowitz understood something his equations could not solve: there is a difference between a model that is **mathematically correct** and the numbers that can actually be **trusted**. The optimiser disappoints not because of its mathematics but because of what we feed it.

Input one • expected returns

Traditionally the historical average — sixty monthly returns, added and divided. The hidden assumption is that sixty observations of a highly variable number give a stable estimate. They do not: one unusually strong quarter can move the average significantly, and ranking stocks by historical average return may be only slightly more useful than ranking them alphabetically.

Machine learning does not remove the noise. It **changes the question**. Instead of asking “what is the average return of this one stock?”, supported by sixty observations, it asks “across thousands of stocks and many years, which characteristics have distinguished stronger from weaker performers?” — a question supported by thousands of observations.

The output must stay honest. No model can claim Reliance will earn exactly 14.2%. A more honest conclusion is that Reliance *ranks* among the stronger group. **Rankings, not exact predictions, are the realistic objective.**

Inputs two and three • correlation and beta

Correlation is traditionally computed from history and then treated as fixed. But a 100-stock portfolio contains **4,950** unique pairs, which no analyst can absorb by reading values — hence the heat map, a colour-coded grid where patterns emerge instantly. More importantly, one fixed grid is not reality: in a panic, correlations rise sharply and assets that normally move independently fall together, precisely when diversification is needed most. A model can classify the market as calm, uncertain or stressed and apply a different correlation map to each.

Beta was calculated from five years of history and treated as fixed. But Reliance in 2015 was an oil and refining business; after expanding into telecommunications it became more connected to consumer technology. The symbol never changed; the business did. A static approach says beta is 1.1. A dynamic approach says it is about 1.18 today, expected to move towards 1.10 next quarter, and that the change is linked to identifiable factors.

THREE INPUTS, THREE IMPROVEMENTS

Input	Traditional method	The hidden assumption	What machine learning changes
Expected returns	Historical average of sixty monthly returns	That sixty observations of a variable number are stable	Asks which characteristics distinguish performers across thousands of stocks — and outputs rankings
Correlation	CORREL on history, then treated as fixed	That the last five years will hold next quarter	Classifies the market regime and applies a different correlation map to each
Beta	Five years of history, treated as fixed	That the business has not changed	Re-estimates continuously and forecasts next quarter with a confidence range

Meera's risk report changes accordingly. Instead of one portfolio beta from older data, it carries **forecast betas for all 80 holdings**, each with a confidence range, highlighting the three whose risk is expected to change most. That is what a risk meeting is for.

And a caution on correlation: in August 2007 several quantitative funds lost heavily within days while the market barely moved. The dangerous correlation was not between stocks but **between strategies** — many funds held similar positions. Modern teams ask: who else owns what we own?

Figure 9.1 The three inputs of Weeks 5 and 6, each improved by changing the question rather than the mathematics.

KEY IDEA

We have not replaced finance with artificial intelligence. The concepts from Weeks 5 and 6 are the foundation of modern machine learning applications in finance. What has been added is richer data, better forecasting and greater computing power — supporting judgement rather than replacing it.

EXAM POINTER

For each of the three inputs, state the traditional method, the hidden assumption and the improvement. The highest-value point is that machine learning changes the *question* rather than removing the noise — and that rankings, not point predictions, are the honest output.

COMMON PITFALL

Being impressed that a model picked the best-performing stock three years running. Ask instead how well it ranked *all* the stocks across many years, using data it had never seen. Three successes may be luck — recall the coin-flipping managers of Topic 2.

QUICK RECAP • TOPIC 9

- Markowitz did not use his own optimiser; the mathematics was correct but the estimates were not trustworthy.
- Historical average returns are too noisy to rank stocks reliably.
- Machine learning changes the question and outputs rankings rather than point predictions.
- Correlations are not fixed — they rise in a panic, exactly when diversification is needed.
- Beta drifts as a business changes; it should be forecast with a confidence range, not stamped on a stock.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Explain why classical models are small and describe sampling noise	Topics 1 and 2 • Figures 1.1, 2.1
2	Apply Bayes in counts and read regression as machine learning	Topics 3, 4 and 5 • Figures 3.1, 4.1, 5.1
3	Distinguish the three families and describe data sources and techniques	Topics 6, 7 and 8 • Figures 6.1, 7.1, 8.1
4	Explain how expected returns, correlation and beta can be improved	Topic 9 • Figure 9.1

The module in six sentences

CAPM uses three inputs because 1960s computing made more impossible, not because more would not help, and machine learning simply asks why useful information should be discarded. Every number computed in this course is an estimate equal to the truth plus sampling noise, and because standard error falls only with the square root of sample size, halving it requires four times the data. Bayes' theorem disciplines what evidence is worth: a five-year winning record lifts the probability of genuine skill from 5% to only about 50%, and a model 90% accurate at spotting rare crashes is still wrong five times out of six. Linear regression is itself a machine learning model — beta is its slope, alpha its intercept, and the residual is unsystematic risk — which means the vocabulary of features, weights and training describes tools already in use. Three families organise the field, and a short ladder of techniques covers most of what is actually done, with regularisation defending against the overfitting that ruins models built on noisy inputs. And the three inputs underpinning Weeks 5 and 6 can each be improved — expected returns becoming rankings drawn from thousands of stocks, correlations becoming condition-dependent, and beta becoming a forecast with a confidence range.

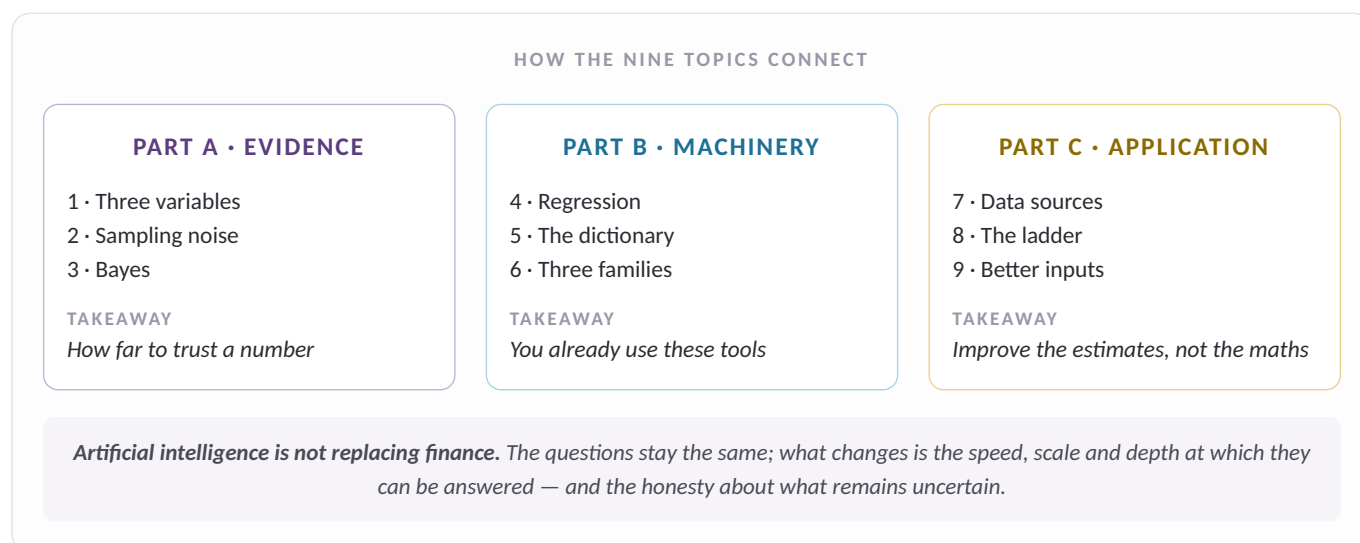


Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check



Term	Definition
Inferential statistics	Drawing conclusions about a large unseen population from a smaller observed sample.
Population	The complete but unobservable set of outcomes.
Sample	The observed subset actually available — for instance sixty monthly returns.
Estimate	A value calculated from a sample; the true value plus sampling noise.
Standard error	The usual amount by which an estimate fluctuates between samples.
Hypothesis test	A formal method for judging whether a pattern is real or could have arisen by chance.
Prior	What was believed before new evidence arrived.
Likelihood	How probable the evidence would be under each possible explanation.
Posterior	The updated belief after weighing the evidence.
Residual	The vertical distance from a data point to the regression line; the unexplained part.
Least squares	The method selecting the line that minimises the sum of squared residuals.
R-squared	The proportion of the target's movement explained by the model.
Multiple regression	Regression using several input variables at once.
Target, or label	The machine learning name for the dependent variable.
Features	The machine learning name for independent variables.
Training	Estimating the relationship between features and target from historical data.
Weights	The machine learning name for regression coefficients.
Overfitting	Memorising historical data rather than learning a pattern that persists; data mining.
Supervised learning	Learning from past examples where the answer is known.
Unsupervised learning	Finding hidden structure in data with no answers attached.
Clustering	Grouping items by similarity without being given categories.
Reinforcement learning	Learning a sequence of decisions through trial, reward and penalty.
Regularisation	Charging a cost for model complexity, to curb confidence in accidental patterns.
Structured data	Information arranged in rows and columns.

Term	Definition
Unstructured data	News, reports, calls and images — information not naturally tabular.
Sentiment score	A number from -1 to +1 representing how positive or negative a text appears.
API	An interface letting one programme request data from another system automatically.
Heat map	A colour-coded grid displaying many correlations at once.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 You want to halve the standard error of an estimated average return. How much more data do you need, and why might collecting it not help?

Four times as much. Standard error falls with the square root of sample size. But extending the window backwards may pull in observations from a company and economy that no longer exist — the Reliance of 2010 is not the Reliance of today, so older data can add noise rather than precision.

Q2 A fund manager has beaten the market five years running. What is the probability of genuine skill, given that 5% of managers are skilled and skilled ones win 9 years in 10?

About 50%. Of 500 managers, 25 are skilled and $25 \times 0.9^5 \approx 15$ of them win five straight. Of the 475 unskilled, $475 \times (1 \div 32) \approx 15$ also do, by luck. About 30 managers show the record; half are genuinely skilled. The prior of 5% becomes a posterior of 50% — a large update, but not certainty.

Q3 A crash model is 90% accurate. Crashes occur in 2% of months. It warns this month. Should you sell?

Investigate, do not sell automatically — the warning is right only about 16% of the time. Across 1,000 months: 18 correct warnings from 20 crashes, and 98 false warnings from 980 calm months. $18 \div 116 \approx 16\%$. When events are rare, false alarms dominate even accurate models.

Q4 A regression gives alpha 0.2% and beta 1.3. The market rises 2% and the stock returns 4%. What was predicted, and what does the gap represent?

Predicted 2.8%; the 1.2% gap is unsystematic risk. $0.2\% + (1.3 \times 2\%) = 2.8\%$. The residual is company-specific information the market return could not explain — exactly the risk diversification removes.

Q5 Classify these: forecasting next quarter's beta; grouping 200 stocks by behaviour; splitting a large order across the trading day.

Supervised, unsupervised, reinforcement. A number predicted from past examples with known answers is supervised. Discovering groups without given categories is unsupervised. A sequence of decisions learned through feedback and reward is reinforcement learning.

Q6 Why can machine learning not simply produce a more accurate expected return for Reliance?

Because it cannot remove sampling noise — it can only change the question. Instead of averaging sixty observations of one stock, it asks which characteristics have distinguished performers across thousands of stocks and many years. The honest output is a *ranking*, not a claim that Reliance will earn exactly 14.2%.